# Technology Exploration Report

SEO Fixer · 30 October 2026 · Sohan Reddy

## What this is

SEO Fixer is a local Django web app with one page. You give it a website URL and, optionally, the site's GitHub repo. It checks the page for SEO problems, asks four AI agents for fixes, and opens a draft pull request with the fixes you pick.

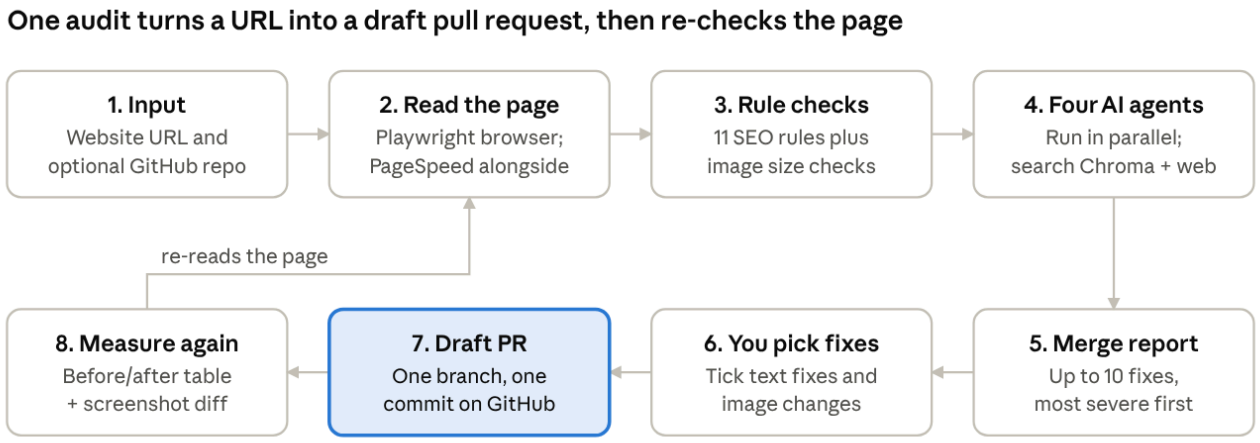
The goal is a terminal app built with React Ink, in custom colors, so a developer can do all of this without leaving the IDE. This report covers what the code uses today and what an Ink app can reuse. Everything here comes from the code in the repo.

## Technologies used today

The app is written entirely in Python and uses 8 packages from `requirements.txt` plus 3 outside services.

| Technology | Version | What it does here |
|---|---|---|
| Django | 5.2.18 | Serves the one page at `/`, handles forms, sessions and CSRF protection |
| SQLite | built in | Stores sessions in `db.sqlite3` (the current audit lives here) |
| Playwright (Chromium) | 1.64.1 | Opens the site in a hidden browser at 1280×800, scrolls it, measures images, takes a full-page screenshot |
| Beautiful Soup | 4.15.0 | Reads the HTML: title, meta tags, headings, links, alt text |
| Anthropic SDK | 2.3.0 | Runs the 4 AI agents (default model `claude-haiku-5-5`) |
| ChromaDB | 1.5.9 | Local search index of 12 Google SEO guide pages, cut into 130-word chunks |
| Pillow | 13.0.0 | Shrinks images and paints changed pixels red in the before/after screenshot |
| Requests | 2.34.2 | Downloads guide pages and calls GitHub and PageSpeed |
| python-dotenv | 1.2.4 | Loads keys from `.env` |
| GitHub REST API | service | Creates a branch, one commit and a draft PR (needs `GITHUB_TOKEN`) |
| Google PageSpeed API | service | Adds 4 Google scores, page weight and load time (needs `PAGESPEED_API_KEY`) |
| Anthropic web search | service | Lets agents search 5 trusted sites for extra guidance |

## How the pieces connect




The **Audit page** button runs steps 1 to 5, **Open draft pull request** runs 6 and 7, and **Measure again** runs step 8. Steps 1 to 3 run with no keys at all.

## The terminal app (Ink)

The audit logic is already separate from the web page, so the Ink app reuses it directly. Ink runs on Node.js and talks to the Python code through a small JSON layer.

| CLI feature | Code it uses | How Ink connects |
|---|---|---|
| Start an audit | `start_audit(url, repo)` in `views.py`, which returns plain data | Calls it through the JSON layer |
| Pick fixes and open a PR | `open_selected_pr(state, selection, include_images)` | Calls it through the JSON layer |
| Measure again | `measurements.measure(url)` and `compare(before, after)` | Calls them through the JSON layer |
| Build the guide index | `ai.ingest()` | Calls it through the JSON layer |
| Keep the current audit | The audit is one plain dictionary | Holds it in the app's state |
| Before/after screenshot | Percent of pixels changed and page height change | Shows them as text in the terminal |

Two connection options, both reusing the functions above:

- A JSON endpoint in Django next to `home`, which Ink calls over HTTP.
- A Django command that prints JSON, which Ink runs as a child process.

Custom colors map to values the code already returns:

- `severity`: `high`, `medium` or `low` on every issue and fix
- `verdict`: `better` or `worse` on every before/after row
- `score`: 0 to 100 for each agent's area
- `notice`: status messages, such as how many guide sources were refreshed

## Design choices

- **Local first.** The app runs on each developer's machine, which suits a workflow that stays inside the IDE.
- **Safe edits.** A text fix is applied only when its exact text appears once in the repo. All changes go into a draft PR for review.
- **Grounded AI.** Agents cite Google's own guidance, and every source and edit is checked before use.
- **Works with zero keys.** Rule checks and measurements always run. AI fixes, PRs and Google scores each turn on with one key.
- **Fast, parallel work.** Google PageSpeed runs alongside the browser, and the 4 agents run at the same time.
- **Sized for typical sites.** Supports repos with up to 150 editable files and measures up to 60 images per page.